

Experiment 'p_5_9_d_cylinder_surface' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_5_9_d_cylinder_surface

Natural language statement: Write a method that computes the surface of a cylinder with a circular base, given radius and height.

Method signature: p_5_9_d_cylinder_surface(r: real, h: real) returns (surface: real)

Ensures

- `surface == 2.0 * 3.14159 * r * h + 2.0 * 3.14159 * r * r`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that computes the surface of a cylinder with a circular base, given radius
→ and height.

The signature should be:

`p_5_9_d_cylinder_surface(r: real, h: real) returns (surface: real)`

The method should respect the following contract:

`ensures surface == 2.0 * 3.14159 * r * h + 2.0 * 3.14159 * r * r`

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-Dn4ZrvV9CBp35T1ZCIMu93CCg2dQP

Model: gpt-5.5-2026-04-23

Created at: 1780588039

Finish reason: stop

Usage: 330 tokens in, and 128 tokens out

```
{  
  surface := 2.0 * 3.14159 * r * h + 2.0 * 3.14159 * r * r;  
}
```

Final Program

```
method p_5_9_d_cylinder_surface(r: real, h: real) returns (surface: real)  
  ensures surface == 2.0 * 3.14159 * r * h + 2.0 * 3.14159 * r * r  
{  
  surface := 2.0 * 3.14159 * r * h + 2.0 * 3.14159 * r * r;  
}
```

Total Token Usage

Input tokens: 330

Output tokens: 128

Reasoning tokens: 74

Sum of ‘total tokens’: 458

Experiment Timings

Overall Experiment started at 1780588039091, ended at 1780588058060, lasting 18969ms (18.97 seconds)

Iteration #1 started at 1780588039093, ended at 1780588058057, lasting 18964ms (18.96 seconds)